

DC-DC BOOST CONVERTER PROTOTYPE

For PV-Integrated Electric Vehicle (Scale-Down Prototype)

20W Solar Panel | 48V Battery Output | DC-DC Boost Converter Design

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1. INTRODUCTION

This report consists of design and development of a DC-DC Boost Converter prototype for a PV-Integrated Electric Vehicle (EV) scale-down prototype.

The system steps up the 20W solar panel output voltage ($V_{mp} = 17.93 \text{ V} - 18.5 \text{ V}$) to charge a 48V battery.

The report covers system specifications, solar panel electrical specifications, design calculations, and component selection.

Battery Ah capacity is not specified at this stage; the laboratory prototype will first verify voltage conversion.

2. SOLAR PANEL ELECTRICAL SPECIFICATIONS (20W)

The following specifications are taken from the 20W solar panel selected for this prototype:

Parameter	Symbol	Value
Maximum Power	P_{ma}^x	20 W
Open Circuit Voltage	V_{ox}	21.4 V
Maximum Power Point Voltage	V_{mp}	17.93 V
Short Circuit Current	I_{sx}	1.22 A
Maximum Power Current	I_{mp}	1.11 A
Module Efficiency	η	15% – 22%
Converter Type	—	DC-DC Boost Converter

3. SYSTEM SPECIFICATIONS

Parameter	Value
PV Panel Power	20 W
Open Circuit Voltage (V_{ox})	21.4 V
Maximum Power Voltage (V_{mp})	17.93 V ($V_i^n = 17.93 \text{ V}$ used in design)
Output Voltage (V_{out})	48 V (Battery Charging)
Battery	48V (Ah to be confirmed after lab testing)
Switching Frequency	20 kHz
Converter Type	DC-DC Boost Converter

4. DESIGN CALCULATIONS

The operating parameters are:

$$V_{in} = 17.93 \text{ V} \mid V_{out} = 48 \text{ V} \mid P_{out} = 100 \text{ W} \mid f = 20 \text{ kHz}$$

4.1 Duty Cycle

$$\begin{aligned} D_{max} &= 1 - \frac{V_{in(min)}}{V_{out}} \\ D_{max} &= 1 - \frac{17.93}{48} \\ D_{max} &= 0.6265 \end{aligned}$$

Required Duty Cycle = 62.65%.

4.2 Input Current

Assuming ideal converter operation with unity efficiency:

$$\begin{aligned} I_{in(max)} &= \frac{20}{18} \\ &= 1.11 \text{ A} \end{aligned}$$

This value matches the panel's I_{mp} range of 1.11 A, confirming the design operates at the maximum power point.

4.3 Output Current

$$\begin{aligned} I_{out} &= \frac{P_{out}}{V_{out}} \\ &= 20 / 48 \\ &= 0.417 \text{ A} \end{aligned}$$

4.4 Inductor Design

The inductor is the energy storage element in the boost converter. From the state equations of the equivalent circuit during the on-state of the switch, the inductor voltage-current relationship governs the ripple:

$$L_p \cdot di^{nd}(t) / dt = V_i^n - R_{pii}^{nd}(t) - V_{out}$$

Assuming ripple current $\Delta I_L = 30\%$ of input current:

$$\Delta I_L = 0.30 \times 1.11 = 0.333 \text{ A}$$

The minimum inductance required to maintain Continuous Conduction Mode (CCM) is:

$$L_{min} = \frac{V_{in(min)} \cdot D_{max}}{\Delta I_L \cdot f_{sw}}$$

$$L_{min} = \frac{17.93 \times 0.6265}{0.333 \times 20000}$$

$$\approx 1683 \text{ } \mu\text{H}$$

Selected value: L = 2200 μ H (toroidal inductor, rated $\geq 2\text{A}$). A value higher than the calculated minimum is selected to sure this is in CCM and reduce current ripple, improving stability during lab testing.

4.5 Output Capacitor

The output capacitor smooths the output voltage ripple. From the capacitor current equation during the switch-off state :

$$C_{p2} \cdot \frac{dV_{out}(t)}{dt} = I_{ind}(t) - I_{load}(t)$$

$$I_{out(max)} = \frac{100}{48}$$

$$\approx 0.417 \text{ A}$$

Allowable output voltage ripple (1% of V_{out}):

$$\Delta V_{out} = 0.01 \times 48 = 0.48 \text{ V}$$

$$C = \frac{I_{out(max)} \cdot D_{max}}{f_{sw} \cdot \Delta V_{out}}$$

$$C = \frac{2.08 \times 0.6265}{20000 \times 0.48}$$

$$= 27.2 \text{ } \mu\text{F}$$

Selected value: C = 47 μ F / 100V electrolytic capacitor

4.6 Load Resistance (Laboratory Testing)

For initial lab testing before battery connection, an equivalent resistive load is used:

$$R_{load} = V_{out}^2 / P_{out} = 48^2 / 20 = 115.2 \text{ } \Omega$$

Selected practical value: 120 Ω / 25W wirewound resistor.

4.7 Summary of Calculated Values

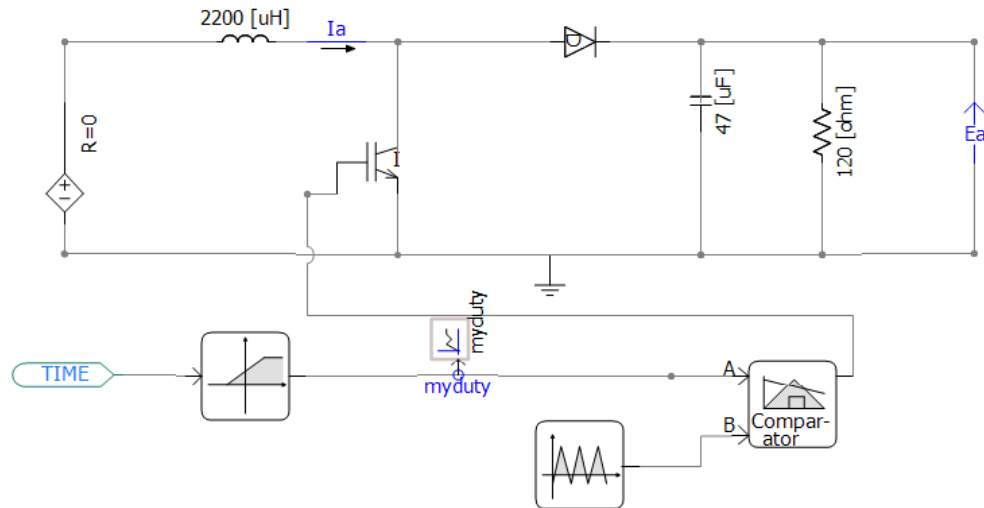
Parameter	Calculated	Selected Value
Duty Cycle (D)	0.6265	62.65%
Input Current	1.11 A	matches I_{mp}
Output Current	0.417 A	—
Inductance (L)	1683 μ H	2200 μ H (toroidal, ≥ 2 A)
Capacitance (C)	27.2 μ F	47 μ F / 100V
Load Resistance	115.2 Ω	120 Ω / 25W wirewound

5. MAIN COMPONENTS SELECTED

Component	Selected Part	Specification / Reason
Solar Panel	100W Poly/Monocrystalline	$V_{mp}=18$ V, $I_{mp}=5.5$ A, $P=20$ W
MOSFET (Switch)	IRFP250N / IRF3205	Rated 200V/30A, low $R_{ds(on)}$
Diode	MBR20100CT Schottky	20A / 100V, fast switching, low V_f
Inductor	2.2 mH Toroidal	Rated ≥ 2 A, ferrite core, ensures CCM
Output Capacitor	47 μ F / 100V Electrolytic	Smooth 48V output, 100V safety margin
Gate Driver	IR2110 / TLP250	Drives MOSFET gate from MCU PWM signal
PWM Controller	Arduino Uno / STM32	Generates 20 kHz PWM at 62.65% duty cycle
Load Resistor (Lab)	120 Ω / 25W Wirewound	Initial testing before battery connection

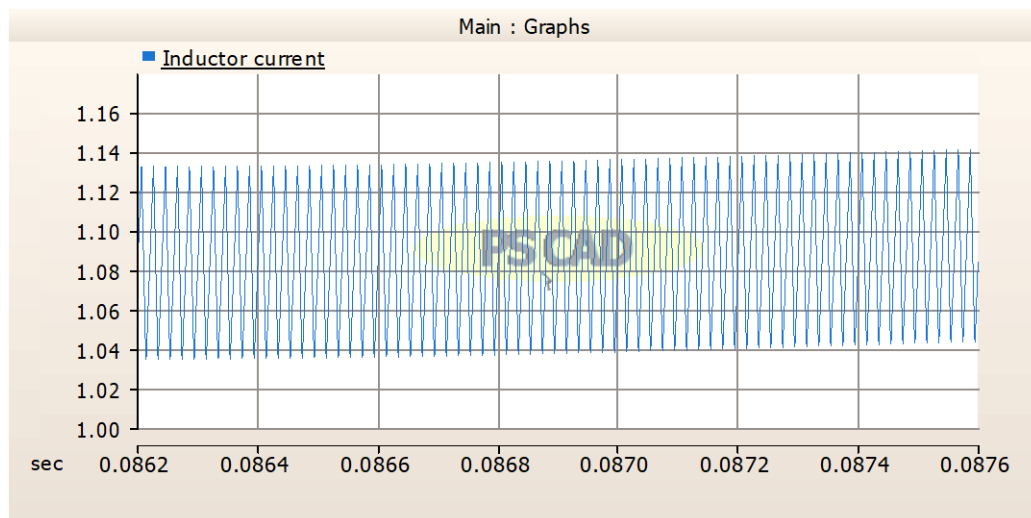
6. BOOST CONVERTER CIRCUIT TOPOLOGY (PSCAD Reference)

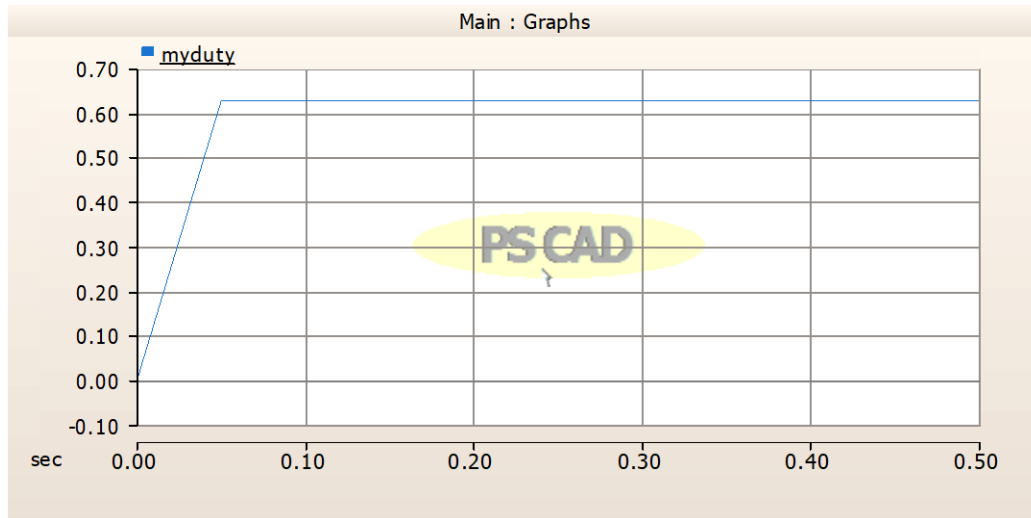
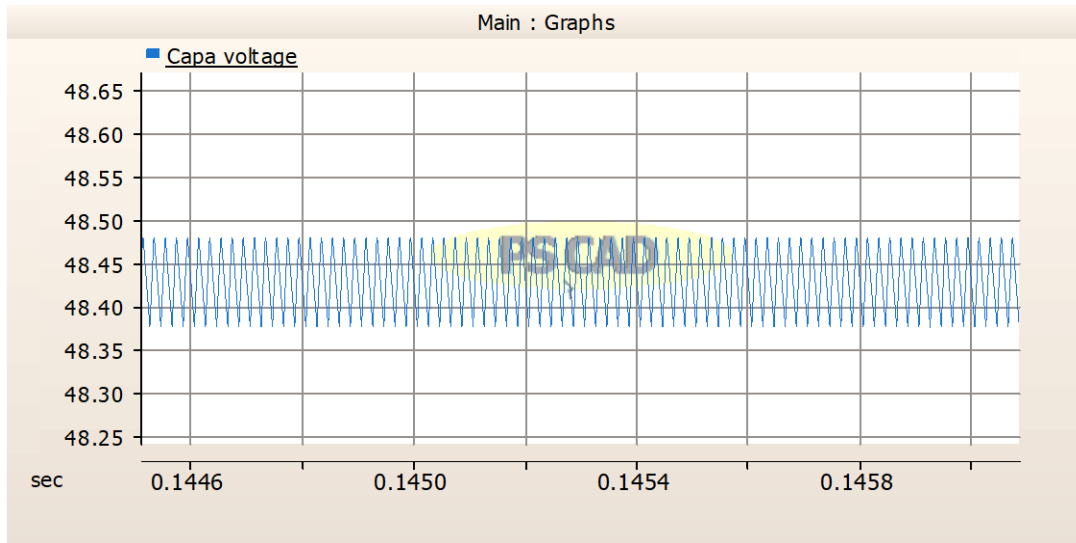
The following represents the boost converter topology for PSCAD simulation and laboratory implementation.



6.1 SIMULATION RESULTS

After resolving a few simulation issues like, incorrect comparator settings and input source modelling, the correct results were obtained. The output voltage settled at 48.4 V which is very close to the designed 48 V target, and the inductor current averaged 1.09 A confirming CCM operation. Overall the simulation results closely match the theoretical design values, validating the boost converter design.





Parameter	Designed	Simulated
Output Voltage	48 V	48.4 V
Inductor Current (avg)	1.11 A	~1.09 A
Current Ripple ΔI_L	0.333 A	~0.10 A
CCM Confirmed	Yes	Yes
Settling Time	< 0.1 s	~0.04 s
Switching Frequency	20 kHz	20 kHz

6.2 PROBLEMS AND SOLUTIONS

Problem	Root Cause	Solution
Vout = 21V (no boosting)	Comparator set to 'Pulse' type instead of 'Level'	Changed comparator to Level type
Vout = 55V instead of 48V	Battery model used as input —dynamic EMF raised effective Vin	Replace with ideal DC voltage source (17.93V)
Inductor current near zero	MOSFET not switching due to comparator issue	Fixed with comparator type correction
Slow settling	Simulation duration too short (0.07s)	Extended duration to 0.3s
Timestep too large	Default timestep cannot resolve 20kHz switching	Set timestep to 0.5 μ s

7. REFERENCES

1. K. Billings, Switchmode Power Supply Handbook, McGraw-Hill, 2nd Edition.
2. P.J. Binduhewa, “Microsource Interface for a Microgrid,” Ph.D. Thesis, University of Manchester, School of Electrical and Electronic Engineering, 2010.